

Normal Distribution The density of the normal random variable X , with mean μ and variance σ^2 , is

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty,$$

where $\pi = 3.14159\dots$ and $e = 2.71828\dots$

The distribution of a normal random variable with mean 0 and variance 1 is called a **standard normal distribution**.

$$Z = \frac{X - \mu}{\sigma}.$$

Example 6.5: Given that X has a normal distribution with $\mu = 300$ and $\sigma = 50$, find the probability that X assumes a value greater than 362.

Solution: The normal probability distribution with the desired area shaded is shown in Figure 6.12. To find $P(X > 362)$, we need to evaluate the area under the normal curve to the right of $x = 362$. This can be done by transforming $x = 362$ to the corresponding z value, obtaining the area to the left of z from Table A.3, and then subtracting this area from 1. We find that

$$z = \frac{362 - 300}{50} = 1.24.$$

Hence,

$$P(X > 362) = P(Z > 1.24) = 1 - P(Z < 1.24) = 1 - 0.8925 = 0.1075.$$



Feature Engineering

- Standardization:** Transforming data to a standard normal distribution (z-scores) improves performance in algorithms sensitive to feature scales.
- Log-Transformations:** Used to make skewed data more normally distributed for better interpretability and model performance.

6.13 A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live

- (a) more than 32 months;
- (b) less than 28 months;
- (c) between 37 and 49 months.

(a) Probability that a mouse lives more than 32 months

1. Compute the z-score:

$$Z = \frac{X - \mu}{\sigma} = \frac{32 - 40}{6.3} = \frac{-8}{6.3} \approx -1.27$$

2. Find the cumulative probability for $Z = -1.27$ from the standard normal table or using a function:

$$P(Z \leq -1.27) \approx 0.102$$

3. The probability of living more than 32 months is:

$$P(X > 32) = 1 - P(Z \leq -1.27) = 1 - 0.102 = 0.898$$

(b) Probability that a mouse lives less than 28 months

1. Compute the z-score:

$$Z = \frac{X - \mu}{\sigma} = \frac{28 - 40}{6.3} = \frac{-12}{6.3} \approx -1.90$$

2. Find the cumulative probability for $Z = -1.90$:

$$P(Z \leq -1.90) \approx 0.028$$

3. Therefore, the probability of living less than 28 months is:

$$P(X < 28) = 0.028$$

(c) Probability that a mouse lives between 37 and 49 months

1. Compute the z-scores for both values:

- For $X = 37$:

$$Z = \frac{37 - 40}{6.3} = \frac{-3}{6.3} \approx -0.48$$

- For $X = 49$:

$$Z = \frac{49 - 40}{6.3} = \frac{9}{6.3} \approx 1.43$$

2. Find the cumulative probabilities for both z-scores:

- For $Z = -0.48$:

$$P(Z \leq -0.48) \approx 0.315$$

- For $Z = 1.43$:

$$P(Z \leq 1.43) \approx 0.921$$

- For $Z = 1.43$:

$$P(Z \leq 1.43) \approx 0.921$$

3. The probability of living between 37 and 49 months is:

$$P(37 \leq X \leq 49) = P(Z \leq 1.43) - P(Z \leq -0.48) = 0.921 - 0.315 = 0.606$$

Example 6.8: An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

Solution: The distribution of light bulb life is illustrated in Figure 6.15. The z values corresponding to $x_1 = 778$ and $x_2 = 834$ are

$$z_1 = \frac{778 - 800}{40} = -0.55 \text{ and } z_2 = \frac{834 - 800}{40} = 0.85.$$

Hence,

$$\begin{aligned} P(778 < X < 834) &= P(-0.55 < Z < 0.85) = P(Z < 0.85) - P(Z < -0.55) \\ &= 0.8023 - 0.2912 = 0.5111. \end{aligned}$$



A binomial random variable can also be approximated by using normal random variable methods discussed above. This approximation can take place as long as:

1. The population size must be **at least** 10 times the sample size.

Normal Approximation to the Binomial Distribution

Let X be a binomial random variable with parameters n and p . For large n , X has approximately a normal distribution with $\mu = np$ and $\sigma^2 = npq = np(1-p)$ and

$$\begin{aligned}P(X \leq x) &= \sum_{k=0}^x b(k; n, p) \\&\approx \text{area under normal curve to the left of } x + 0.5 \\&= P\left(Z \leq \frac{x + 0.5 - np}{\sqrt{npq}}\right),\end{aligned}$$

and the approximation will be good if np and $n(1-p)$ are greater than or equal to 5.

Example 6.15: The probability that a patient recovers from a rare blood disease is 0.4. If 100 people are known to have contracted this disease, what is the probability that fewer than 30 survive?

Solution: Let the binomial variable X represent the number of patients who survive. Since $n = 100$, we should obtain fairly accurate results using the normal-curve approximation with

$$\mu = np = (100)(0.4) = 40 \text{ and } \sigma = \sqrt{npq} = \sqrt{(100)(0.4)(0.6)} = 4.899.$$

To obtain the desired probability, we have to find the area to the left of $x = 29.5$.

The z value corresponding to 29.5 is

$$z = \frac{29.5 - 40}{4.899} = -2.14,$$

and the probability of fewer than 30 of the 100 patients surviving is given by the shaded region in Figure 6.26. Hence,

$$P(X < 30) \approx P(Z < -2.14) = 0.0162.$$



Example 6.16: A multiple-choice quiz has 200 questions, each with 4 possible answers of which only 1 is correct. What is the probability that sheer guesswork yields from 25 to 30 correct answers for the 80 of the 200 problems about which the student has no knowledge?

Solution: The probability of guessing a correct answer for each of the 80 questions is $p = 1/4$. If X represents the number of correct answers resulting from guesswork, then

$$P(25 \leq X \leq 30) = \sum_{x=25}^{30} b(x; 80, 1/4).$$

Using the normal curve approximation with

$$\mu = np = (80) \left(\frac{1}{4} \right) = 20$$

and

$$\sigma = \sqrt{npq} = \sqrt{(80)(1/4)(3/4)} = 3.873,$$

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we need the area between $x_1 = 24.5$ and $x_2 = 30.5$. The corresponding z values are

$$z_1 = \frac{24.5 - 20}{3.873} = 1.16 \text{ and } z_2 = \frac{30.5 - 20}{3.873} = 2.71.$$

The probability of correctly guessing from 25 to 30 questions is given by the shaded region in Figure 6.27. From Table A.3 we find that

$$\begin{aligned} P(25 \leq X \leq 30) &= \sum_{x=25}^{30} b(x; 80, 0.25) \approx P(1.16 < Z < 2.71) \\ &= P(Z < 2.71) - P(Z < 1.16) = 0.9966 - 0.8770 = 0.1196. \end{aligned}$$



Exponential Distribution The continuous random variable X has an **exponential distribution**, with parameter β , if its density function is given by

$$f(x; \beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\beta > 0$.

Example 6.18: Suppose that telephone calls arriving at a particular switchboard follow a Poisson process with an average of 5 calls coming per minute. What is the probability that up to a minute will elapse by the time 2 calls have come in to the switchboard?

Solution: The Poisson process applies, with time until 2 Poisson events following a gamma distribution with $\beta = 1/5$ and $\alpha = 2$. Denote by X the time in minutes that transpires before 2 calls come. The required probability is given by

$$P(X \leq 1) = \int_0^1 \frac{1}{\beta^2} x e^{-x/\beta} dx = 25 \int_0^1 x e^{-5x} dx = 1 - e^{-5}(1 + 5) = 0.96. \quad \blacksquare$$

While the origin of the gamma distribution deals in time (or space) until the occurrence of α Poisson events, there are many instances where a gamma distribution works very well even though there is no clear Poisson structure. This is particularly true for **survival time** problems in both engineering and biomedical applications.

